OmniScope: Modality-Decoupled Token Compression for Omnimodal Large Language Models

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Abstract

Existing token compression methods for omnimodal large language models typically rely on one modality to determine what to retain in the other. We show that this assumption often breaks down: for the same query, audio and video relevance often peaks at different moments. This cross-modal salience mismatch makes unidirectional guidance prone to discarding answer-critical cues under aggressive compression. We propose OmniScope, a training-free token compression framework that uses the query as a shared semantic anchor while estimating relevance separately for audio and video. OmniScope allocates modality-specific token budgets, prunes visual tokens with an anchor-delta strategy that preserves both global context and temporal changes, and merges audio tokens within each second to reduce redundancy while maintaining temporal continuity. Across four audio-video benchmarks and two Qwen2.5-Omni model scales, OmniScope achieves the best average accuracy across all compression settings. At 25% overall token retention, it delivers up to 3.53x prefill speedup and more than 15% GPU memory reduction, with only a 0.35-point drop in average accuracy. These results suggest a simple design principle for OmniLLM inference: share the query across modalities, but not the salience estimates.

CCS Concepts

• **Computing methodologies → Computer vision**; *Natural language processing*; Machine learning.

Keywords

Token compression, omnimodal large language models, modality-decoupled compression, audio-visual understanding, inference efficiency

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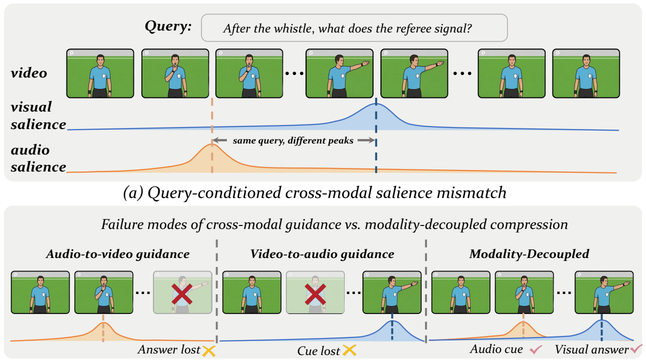
1 Introduction

Omnimodal large language models (OmniLLMs) [9, 21, 30, 32, 39, 41–43, 45, 47] can jointly process visual, auditory, and textual modalities, enabling end-to-end audio-video understanding and representing a major advance in multimodal intelligence. Compared with models that handle only images or text, OmniLLMs capture the synergistic semantics between visual scenes and acoustic signals, delivering significant advantages in applications such as video question answering, content moderation, and real-time interaction.

However, deploying OmniLLMs in practice poses severe efficiency challenges. A typical OmniLLM decomposes a video into visual frames and an audio stream, and encodes them into token sequences via separate encoders. As video duration increases, the total number of audio-visual tokens grows rapidly, incurring substantial computational and memory overhead while severely degrading the quality of contextual interaction, limiting real-time deployment on resource-constrained devices.

Token compression offers a direct means to alleviate this bottleneck. In vision-only multimodal LLMs, a variety of token compression methods have been proposed to reduce inference cost by pruning or merging redundant visual tokens [2, 26, 33, 40, 44]. However, these methods target purely visual settings and cannot directly address the joint compression of audio and video tokens. The few methods designed for omnimodal scenarios, such as OmniZip [34], adopt unidirectional cross-modal guidance strategies, letting one modality determine what is retained in the other. However, in OmniLLMs, the query is shared across modalities, but the relevance pattern is often not—visually decisive moments may be acoustically unremarkable, while audio-critical segments may exhibit little visual change. As illustrated in Fig. 1, this misalignment



(a) Query-conditioned cross-modal salience mismatch

(b) Why explicit cross-modal guidance can fail

Figure 1: One query, two modalities, and often two very different sparsity patterns. (a) Audio salience peaks early (whistle) while visual salience peaks late (hand signal), illustrating cross-modal salience mismatch. (b) Unidirectional cross-modal guidance loses critical information, while modality-decoupled compression preserves both.

of cross-modal salience peaks means that low-salience regions in one modality may carry critical cues for the other, making unidirectional guidance prone to erroneously discarding key information under high compression ratios.

Our analysis of the attention distribution in OmniLLMs further reveals that the consequences of this mismatch are more severe than they may appear: as shown in Fig. 3, token compression significantly strengthens cross-window modal interaction, but when unidirectional guidance discards critical tokens under salience mismatch, the enhanced cross-window attention is instead steered toward secondary cues. Conversely, if each modality retains only its own query-relevant tokens, the amplified cross-window attention naturally falls on task-relevant cues from both sides. This motivates the core design principle of OmniScope: using the query as a shared semantic anchor while letting each modality independently decide its own compression strategy, ensuring that critical information from both modalities is preserved and that the interaction enhancement brought by compression is directed toward task-relevant cues. Building on this principle, we propose OmniScope, a training-free token compression framework for OmniLLMs.

OmniScope uses the query as a shared semantic anchor while allowing audio and visual tokens to independently assess their importance and allocate compression budgets. On this basis, the visual side employs Anchor-Delta Spatio-Temporal Compression (AD-STC) to balance global semantic coverage and temporal increments, while the audio side adopts per-second Token Merging to reduce redundancy while maintaining temporal continuity.

We validate OmniScope on four multimodal video understanding benchmarks at two model scales (7B and 3B). OmniScope achieves the best average performance at all compression ratios, with nearly lossless accuracy at 45% retention. At the more aggressive 25% retention, OmniScope still maintains significantly stronger robustness than all competing methods, while delivering up to 3.53× prefilling speedup and over 15% GPU memory reduction.

Our contributions are summarized as follows:

- We identify that the unidirectional cross-modal guidance assumed by prior omnimodal compression methods is unreliable under cross-modal salience mismatch, and propose OmniScope, a training-free, modality-decoupled token compression framework that uses the query as a shared semantic anchor while allowing each modality to independently compress its tokens without explicit cross-modal guidance.
- We propose AD-STC, an anchor-delta spatio-temporal compression strategy that addresses the loss of global semantic coverage in existing video token compression methods.
- On the audio side, we introduce per-second bipartite soft matching on post-encoder temporal embeddings, addressing the coarse granularity of existing audio compression while preserving temporal continuity.
- On four benchmarks and two model scales, OmniScope achieves the best average performance at all compression ratios, with nearly lossless accuracy at 45% retention.

2 Related Work

2.1 Omnimodal Large Language Models

Omnimodal large language models (OmniLLMs) [9, 21, 30, 32, 39, 41–43, 45, 47] aim to unify visual, auditory, and textual modalities within a single model for end-to-end audio-video understanding. Compared with vision-language models [3, 4, 15, 18, 20, 36] that handle only visual and textual modalities, OmniLLMs capture the synergistic semantics between visual scenes and acoustic signals, delivering significant advantages in applications such as video question answering, content moderation, and real-time interaction. Representative open-source systems include Qwen2.5-Omni [41] and VITA-1.5 [9], while proprietary systems such as GPT-4o [12] and Gemini [35] further demonstrate the effectiveness of this paradigm. These systems typically employ modality-specific encoders and a time-window interleaving mechanism to organize multimodal tokens. As video duration grows, the combined audio-visual token count scales rapidly, making token compression a key problem [13] for improving the inference efficiency of OmniLLMs.

2.2 Token Compression in Multimodal Models

In the visual domain, token compression methods have been extensively studied [28] and mainly include similarity-based token merging [1, 27], query-based text-guided compression [22, 29, 31, 46], and attention-based token pruning [2, 26, 40, 44]. In the video domain, spatio-temporal compression methods [11, 27, 33, 37] further exploit inter-frame redundancy to reduce the token count. However, existing video spatio-temporal compression methods typically apply the same compression strategy to all temporal incremental frames, overlooking the differences in semantic density and temporal variation across frames, which tends to cause a loss of global semantic coverage under high compression ratios. On the audio side, representative token reduction techniques include adjacent token merging [17] and importance-based pruning [14, 19], which are constrained by the encoder's internal structure or coarse-grained compression strategies, making it difficult to flexibly adapt to the compression ratios required by downstream LLMs and to minimize information loss during compression. Recent work has begun to explore audio–visual joint compression. OmniZip [34] is

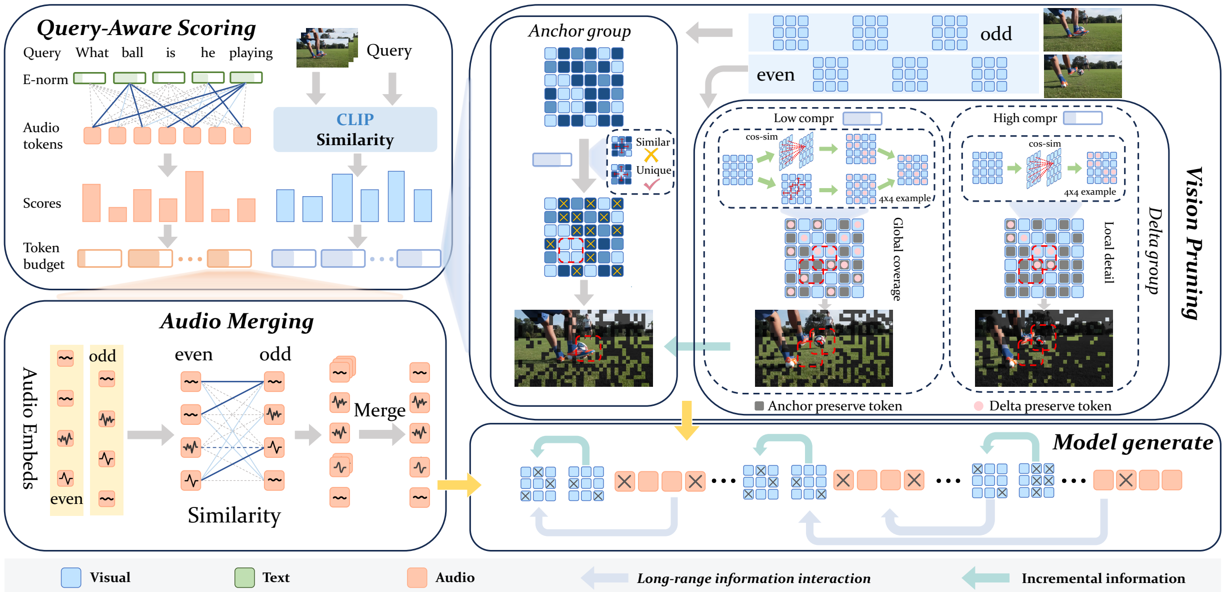


Figure 2: Overview of OmniScope. OmniScope operates in three stages: (1) Query-Aware Scoring independently estimates the importance of visual and audio tokens relative to the query and allocates per-position compression budgets; (2) Vision Pruning balances global semantic coverage and temporal increments through anchor–delta frame alternation; (3) Audio Merging fuses redundant tokens via per-second bipartite soft matching.

a training-free method that uses audio importance scores to guide video token pruning, while OmniSIFT [6] requires end-to-end fine-tuning of both an additional compression module and the LLM decoder, and employs vision-guided modality-asymmetric compression. Although the two methods guide compression in opposite directions, both rely on unidirectional cross-modal guidance—using one modality's importance to determine the retention strategy of the other. Moreover, the training dependency of OmniSIFT limits its generalizability across different models.

3 Method

3.1 Preliminaries

Omnimodal large language models (OmniLLMs) aim to jointly process visual, auditory, and textual modalities for end-to-end audio-video understanding. Taking Qwen2.5-Omni [41] as a representative example, a typical OmniLLM comprises four components: a vision encoder Φ_v, an audio encoder Φ_a, a projector, and an LLM backbone. Given a video, the system first decomposes it into a sequence of video frames X_{vid} and the corresponding audio segments X_{aud}. The vision encoder and audio encoder convert them into token sequences respectively:

$$Z_v = \Phi_v(X_{\text{vid}}) \in \mathbb{R}^{N_v \times D}, \quad Z_a = \Phi_a(X_{\text{aud}}) \in \mathbb{R}^{N_a \times D}, \tag{1}$$

where N_v and N_a denote the number of visual and audio tokens, and D is the embedding dimension. The projector maps both token types into the LLM's embedding space, where they are processed together with text tokens to generate a response.

To organize the audio-video tokens, OmniLLMs adopt a *time-window interleaving* mechanism. Specifically, the audio and video streams are segmented into W time windows of fixed duration. Within each window, co-temporal visual tokens and audio tokens are aligned and concatenated into a cross-modal block; these blocks are then arranged chronologically to form the full input sequence to the LLM. Let the i-th time window contain $n_v^{(i)}$ visual tokens and $n_a^{(i)}$ audio tokens. The sequence received by the LLM can be expressed as:

$$X_{\text{LLM}} = \left[B_1, B_2, \ldots, B_W\right], \quad B_i = \left[Z_v^{(i)}; Z_a^{(i)}\right], \tag{2}$$

where $[\,\cdot\,;\,\cdot\,]$ denotes concatenation along the sequence dimension. Under this mechanism, visual and audio tokens within each window jointly participate in the LLM's self-attention computation. As the video duration increases, the total number of audio-visual tokens grows rapidly, not only incurring substantial computational and memory overhead but also allowing a large volume of redundant tokens to participate in attention computation, posing challenges to both the efficiency and quality of the model's reasoning.

3.2 Motivation: From Attention Dilution to Salience Mismatch

Since tokens within each temporal window jointly participate in softmax-normalized self-attention, redundant tokens within the window can dilute the attention budget available to truly important cross-window tokens. To verify this, we visualize the attention distribution across different layers of Qwen2.5-Omni-3B, as illustrated in Fig. 3 (showing Layer 3 as a representative example). Under the uncompressed Full Token setting, attention is heavily concentrated within the diagonal blocks of local temporal windows, while both cross-window attention and cross-modal attention are extremely weak. After random token compression, the number of

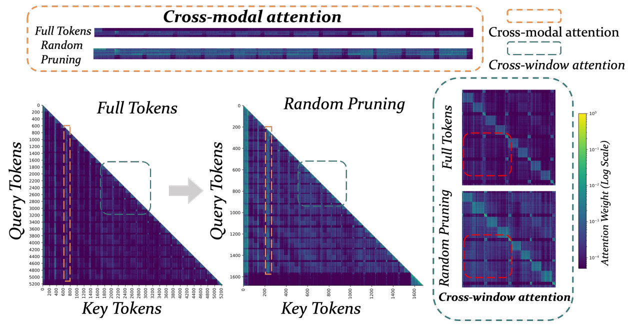


Figure 3: Comparison of attention maps at Layer 3 of Qwen2.5-Omni-3B. Under the Full Token setting, attention is concentrated locally; after random pruning, both cross-window (the off-diagonal regions highlighted by red dashed boxes are notably enhanced) and cross-modal attention are significantly strengthened.

tokens within each window is reduced, and notable attention activation emerges in the off-diagonal cross-window regions, with cross-modal attention similarly strengthened. This finding demonstrates that compression can effectively enhance cross-window modal interaction. Unidirectional cross-modal guidance methods [34] essentially exploit this property: by using one modality as the guide and removing redundant tokens of the other modality around it, these methods strengthen the interaction between the guiding modality's information and other temporal windows, thereby maintaining accuracy after compression. However, since the two modalities often exhibit very different relevance patterns for the same query (as illustrated in Fig. 1), the importance scores of the guiding modality cannot reliably reflect the information value of the other modality. We further quantify the pervasiveness of this mismatch using CLIP and CLAP as external probes: across 1,197 query–video pairs, approximately 78.3% exhibit only weak Spearman correlation ($|\rho| < 0.3$) between visual and audio salience scores. Under high compression ratios, this mismatch causes critical tokens in the non-guiding modality to be erroneously discarded, resulting in the loss of key cross-modal information. This directly inspires the modality-decoupled design of OmniScope: using the query as a shared semantic anchor, each modality independently removes task-irrelevant redundant tokens, so that the retained core tokens cluster around the same task semantics.

3.3 OmniScope

OmniScope is a training-free, query-aware omni-token compression framework consisting of three stages (Fig. 2): query-aware scoring and budget allocation, anchor–delta spatio-temporal vision pruning, and per-second audio token merging. Visual and audio relevance is estimated independently, avoiding explicit cross-modal guidance.

3.3.1 Visual Scoring. To assess the semantic relevance of each video frame to the input query, we adopt CLIP [25] as the visual scorer (Fig. 2, Query-Aware Scoring, right). This scorer operates independently of the OmniLLM's vision encoder, thereby decoupling importance estimation from model inference. Given sampled video frames and a query q, we extract the image feature $\mathbf{f}_I^{(t)} \in \mathbb{R}^{D_c}$ for the t-th frame and the query feature $\mathbf{f}_Q \in \mathbb{R}^{D_c}$, and compute a frame-level importance score via cosine similarity:

$$s_v^{(t)} = \frac{\mathbf{f}_I^{(t)} \cdot \mathbf{f}_Q}{\|\mathbf{f}_I^{(t)}\|_2 \, \|\mathbf{f}_Q\|_2}. \tag{3}$$

3.3.2 Audio Scoring. A natural counterpart on the audio side would be an external cross-modal model such as CLAP [38]. However, models of this family are designed for clip-level audio–text alignment and typically operate at a granularity of several seconds or longer, whereas OmniScope requires per-second or even per-token importance estimation. This coarse temporal resolution makes CLAP ill-suited for the fine-grained scoring needed by our compression framework.

We instead observe that the OmniLLM's audio encoder outputs, after passing through the projector, are already mapped into the same embedding space as the LLM tokens, allowing the relevance between audio tokens and the query text to be computed directly without an external model (Fig. 2, Query-Aware Scoring). In principle, the same strategy could be applied to the visual side, but replacing CLIP with the model's internal visual embeddings reduces average accuracy by 0.47–0.93 points, suggesting insufficient vision–text alignment for reliable scoring; the visual side therefore still relies on the external CLIP scorer.

Concretely, let the audio encoder output be $\mathbf{Z}_a \in \mathbb{R}^{N_a \times D}$ and the query embedding obtained from the LLM's embedding layer be $\mathbf{E}_Q \in \mathbb{R}^{L \times D}$. Prior work has shown that the squared norm of word embeddings encodes the information gain conveyed by the word, with semantically informative words exhibiting larger norms and high-frequency function words exhibiting smaller norms [23]. We therefore use the embedding norm as a simple gating signal to down-weight uninformative tokens:

$$g_j = \frac{\|\mathbf{E}_Q^{(j)}\|}{\max_k \|\mathbf{E}_Q^{(k)}\|}. \tag{4}$$

We compute the gated cross-modal similarity matrix and aggregate it into a per-token importance score:

$$S_{ij} = \bar{\mathbf{z}}_a^{(i)} \cdot \bar{\mathbf{e}}_Q^{(j)},\, w_{ij} = \mathrm{softmax}_j\left(\frac{g_j\, S_{ij}}{\tau}\right),\, s_{\text{audio}}^{(i)} = \textstyle\sum_j\, w_{ij}\, S_{ij}, \tag{5}$$

where $\bar{\cdot}$ denotes ℓ_2-normalization, and i indexes audio tokens while j indexes query tokens. Scores are then aggregated at one-second granularity by averaging the top-k token scores within each second, yielding per-second importance values $s_a^{(s)}$. Since audio signals exhibit strong temporal continuity, the context surrounding a high-scoring second is often equally informative. We therefore apply a decaying score propagation to positions near key seconds. We initialize $\hat{s}_a^{(s)} \leftarrow s_a^{(s)}$ for all s, and let $\mathcal{H}$ denote the set of seconds whose score exceeds the P-th percentile. For each second s, we compute a boost factor from all nearby high-scoring seconds and apply it:

$$b_s = \max_{s' \in \mathcal{H},\, 0<|s-s'|\le R} \left(1 + (\beta - 1) \cdot \gamma^{|s-s'|}\right), \quad \hat{s}_a^{(s)} \leftarrow \hat{s}_a^{(s)} \cdot b_s, \tag{6}$$

where β controls the base boost magnitude, γ is the distance decay rate, and R is the propagation radius.

3.3.3 Token Budget Allocation. Once the importance scores are obtained, each modality is allocated a compression budget independently. Given a modality's global retention ratio $1-\rho$, the total number of retained tokens is $N^{\text{budget}} = (1-\rho) \cdot N_{\text{total}}$. Based on the position-level importance scores $\{s^{(i)}\}$ (frame-level for vision, second-level for audio), the number of tokens retained at each position is allocated proportionally:

$$n_{\text{keep}}^{(i)} = \left\lfloor \frac{s^{(i)}}{\sum_j s^{(j)}} \cdot N^{\text{budget}} + 0.5 \right\rfloor. \tag{7}$$

The compression ratios ρ_v and ρ_a are configured independently—each modality determines its retained token count based solely on its own query relevance, without cross-modal comparison.

3.4 AD-STC

This section addresses *which* tokens to retain within each frame. Existing video token compression methods [11, 27, 33, 37] typically use a uniform strategy for temporal incremental frames, often losing global semantic coverage under high compression. AD-STC instead alternates frames between two complementary roles (Fig. 2, Vision Pruning). *Anchor frames* retain spatially distinctive tokens to capture current semantics, while *delta frames* retain the largest information increments relative to the preceding anchor. Delta scoring adapts to the compression ratio, balancing global coverage and local detail with a generous budget and prioritizing dense, coherent temporal increments with a tight budget. Together, the two roles preserve spatial and temporal information.

3.4.1 Anchor Frame: Spatial Density Selection. Anchor frames aim to retain the most spatially informative tokens within the allocated budget. We adopt a density-peak clustering approach with k-nearest neighbors (DPC-KNN) [7]. For each token $\mathbf{h}_i$ in the frame, we compute its local density as the mean similarity to its k nearest neighbors:

$$d_i = \frac{1}{k} \sum_{\mathbf{h}_j \in \text{KNN}(\mathbf{h}_i)} \text{sim}(\mathbf{h}_i, \mathbf{h}_j). \tag{8}$$

A high d_i indicates that the token closely resembles its neighbors and is therefore spatially redundant, while a low d_i indicates that it is relatively isolated in the feature space and carries distinctive semantic content. We rank all tokens by d_i in ascending order and retain the $n_{\text{keep}}^{(t)}$ tokens with the lowest values.

3.4.2 Delta Frame: Adaptive Temporal Selection. Delta frames aim to capture the information increment relative to the preceding anchor frame. We adaptively switch between two scoring strategies according to the retention ratio $r = n_{\text{keep}}^{(t)}/n_{\text{total}}$: when $r \geq \tau_r$, a joint spatio-temporal criterion is used; when $r < \tau_r$, a temporal-difference criterion is applied instead.

Low compression (generous budget). When the retention ratio is high, we employ a joint irreplaceability score that accounts for both spatial and temporal dimensions simultaneously. Let $\{\mathbf{h}_i\}$ denote the tokens of the current delta frame and $\{\mathbf{h}_k^{\text{prev}}\}$ denote the tokens of the preceding anchor frame. For each token $\mathbf{h}_i$, we compute a spatial replaceability term and a temporal replaceability term:

$$\alpha_i^{\text{intra}} = \max_{j \neq i} \text{sim}(\mathbf{h}_i, \mathbf{h}_j), \quad \alpha_i^{\text{inter}} = \max_{l} \text{sim}(\mathbf{h}_i, \mathbf{h}_l^{\text{prev}}), \tag{9}$$

where α_i^{intra} measures how well a token can be substituted by another token in the same frame (a high value indicates spatial redundancy), and α_i^{inter} measures how well it can be substituted by a token in the preceding anchor frame (a high value indicates temporal redundancy).The two terms are fused multiplicatively into an overall replaceability score:

$$\text{replaceability}_i = \alpha_i^{\text{intra}} \times \alpha_i^{\text{inter}}. \tag{10}$$

A token has a high product only when it is replaceable in both dimensions and is therefore removed; one replaceable in only one dimension (e.g., spatially redundant but temporally novel) receives a low score and is retained. Compared with single-dimension criteria, this multiplicative fusion yields a more dispersed spatio-temporal selection and more uniform global coverage when the budget is sufficient. We retain the $n_{\text{keep}}^{(t)}$ tokens with the lowest replaceability scores.

High compression (tight budget). When the retention ratio is low, the token budget is severely constrained. Under such conditions, the joint spatio-temporal criterion tends to scatter the few retained tokens across both spatial and temporal dimensions, yielding sparse, fragmented information that lacks local coherence. We therefore concentrate the limited budget on capturing dense, locally complete temporal increments: we compute the per-position cosine similarity between the current and the preceding frame, $c_i = \text{sim}(\mathbf{h}_i, \mathbf{h}_i^{\text{prev}})$, and retain the $n_{\text{keep}}^{(t)}$ tokens with the lowest similarity—i.e., those that have changed the most.

3.5 Audio Token Compression

Due to the short-term stationarity of audio signals [24], adjacent tokens are highly redundant yet retain subtle acoustic differences, making direct pruning prone to temporal information gaps. We therefore perform per-second merging on post-encoder temporal embeddings, providing finer-grained compression than existing audio methods [17, 19] while minimizing information loss and preserving temporal coverage (Fig. 2, Audio Merging).

Given the per-second token budgets $\{n_{\text{keep}}^{(s)}\}$ produced by the allocation described in Sec. 3.3, we perform token merging via bipartite soft matching [1] independently within each one-second segment. The N tokens in a segment are first partitioned into two disjoint sets by alternating index: a *source* set $\mathcal{A} = \{\mathbf{h}_0, \mathbf{h}_2, \ldots\}$ (even-indexed) and a *destination* set $\mathcal{B} = \{\mathbf{h}_1, \mathbf{h}_3, \ldots\}$ (odd-indexed). A similarity matrix between the two sets is then computed as:

$$\mathbf{M} = \bar{\mathbf{H}}_{\mathcal{A}} \bar{\mathbf{H}}_{\mathcal{B}}^{\top}, \tag{11}$$

where $\bar{\cdot}$ denotes ℓ_2-normalization. Each source token is paired with its most similar destination token. Among all such pairs, the $m = N - n_{\text{keep}}^{(s)}$ pairs with the highest similarity scores are selected for merging. Each selected source token is fused into its matched destination through averaging:

$$\mathbf{h}'_{\text{dst}} = \frac{\mathbf{h}_{\text{dst}} + \sum_{i \in \mathcal{S}(\text{dst})} \mathbf{h}_i}{1 + |\mathcal{S}(\text{dst})|}, \tag{12}$$

where $\mathcal{S}(\text{dst})$ is the set of source tokens assigned to the destination token. If the required reduction exceeds $\lfloor N/2 \rfloor$, this step is applied iteratively.

Table 1: Comparison on omnimodal (audio & video) QA benchmarks at 45% and 25% retained ratios. The best result among compression methods for each metric is bolded. Δ denotes the difference relative to the Full Tokens baseline.

Method	Ratio	WorldSense	DailyOmni	OmniVideoBench	Video-MME	Avg.	Δ
				Qwen2.5-Omni-7B			
Full Tokens	100%	46.0	62.0	34.1	63.3	51.35	–
Random	45%	44.7	58.7	34.2	63.3	50.23	−1.12
FastV (A&V)	45%	45.8	59.5	**35.8**	63.4	51.13	−0.22
OmniZip	45%	45.7	60.0	35.6	63.3	51.15	−0.20
OmniScope (Ours)	45%	**46.3**	**60.5**	35.5	**64.3**	**51.65**	**+0.30**
Random	25%	44.9	56.9	34.9	62.4	49.78	−1.57
FastV (A&V)	25%	44.7	58.4	35.6	63.4	50.53	−0.82
OmniZip	25%	44.8	57.8	33.7	62.9	49.80	−1.55
OmniScope (Ours)	25%	**45.7**	**58.8**	**35.8**	**63.7**	**51.00**	**−0.35**
				Qwen2.5-Omni-3B			
Full Tokens	100%	46.1	60.7	32.7	61.1	50.15	–
Random	45%	44.1	57.6	32.4	60.9	48.75	−1.40
FastV (A&V)	45%	44.2	**59.7**	33.0	61.2	49.53	−0.62
OmniZip	45%	45.4	59.1	**33.2**	61.1	49.70	−0.45
OmniScope (Ours)	45%	**46.1**	59.3	**33.2**	**62.1**	**50.18**	**+0.03**
Random	25%	42.5	55.7	31.7	59.8	47.43	−2.72
FastV (A&V)	25%	42.9	55.8	32.8	**60.4**	47.98	−2.17
OmniZip	25%	44.0	56.6	**33.1**	59.4	48.28	−1.87
OmniScope (Ours)	25%	**44.5**	**56.9**	**33.1**	60.1	**48.65**	**−1.50**

4 Experiments

4.1 Experimental Setup

Benchmarks. We evaluate OmniScope on four audio-video understanding benchmarks that span diverse task types and video durations: (1) **WorldSense** [10], which assesses joint audio-video comprehension across eight domains—Tech & Science, Culture & Politics, Daily Life, Film & TV, Performance, Games, Sports, and Music; (2) **DailyOmni** [48], which targets omnimodal question answering in everyday scenarios; (3) **OmniVideoBench** [16], a comprehensive benchmark for omnimodal video understanding; and (4) **Video-MME** [8], a widely adopted video understanding benchmark in which incorporating audio further improves accuracy. Together, these benchmarks cover a broad range of video durations, task types, and audio-visual modality balance.

Base Models. Following prior work [34], we implement OmniScope on Qwen2.5-Omni [41] at two parameter scales, 7B and 3B. FlashAttention-2 [5] is enabled in all experiments.

Comparison Methods. Token compression methods designed specifically for OmniLLMs remain scarce. We therefore adapt several representative approaches for comparison. Full Tokens serves as the uncompressed baseline. Random applies equal-ratio random pruning to both modalities. OmniZip [34], the current state-of-the-art open-source training-free omnimodal token compression method, employs audio-guided dynamic video token pruning coupled with an interleaved spatio-temporal compression module. FastV (A&V) [2] performs attention-based token pruning at a shallow decoder layer, applied jointly to audio and video tokens; we adapt the original implementation for FlashAttention compatibility by computing importance scores solely from the attention distribution of the last text token at the designated layer, which also reduces its memory overhead. We do not compare with OmniSIFT [6], as it requires additional training and its code is not publicly available at the time of submission, precluding reproducible inference-time comparison.

Implementation Details. All experiments are conducted on NVIDIA A800 (80 GB) GPUs. Audio scoring leverages the model's own audio encoder. Visual scoring employs CLIP ViT-L/14@336px [25]. All hyperparameters are fixed across all benchmarks and both model scales without per-dataset tuning. We report results under two compression settings, referred to as 45% retention (ρ_v=0.6, ρ_a=0.25) and 25% retention (ρ_v=0.8, ρ_a=0.35), where the percentages denote the approximate overall token retention ratio. Video input is uniformly capped at 128 frames across all benchmarks.

4.2 Main Results

State-of-the-Art Compression Performance. As shown in Table 1, we evaluate OmniScope on four omnimodal audio-video understanding benchmarks under 45% and 25% token retention ratios. Across all settings, OmniScope consistently achieves the highest average accuracy among compression methods, with nearly lossless accuracy at 45% retention. Under the more aggressive 25% retention setting, OmniScope incurs the smallest accuracy drop relative to the Full Tokens baseline at both model scales. For example, on

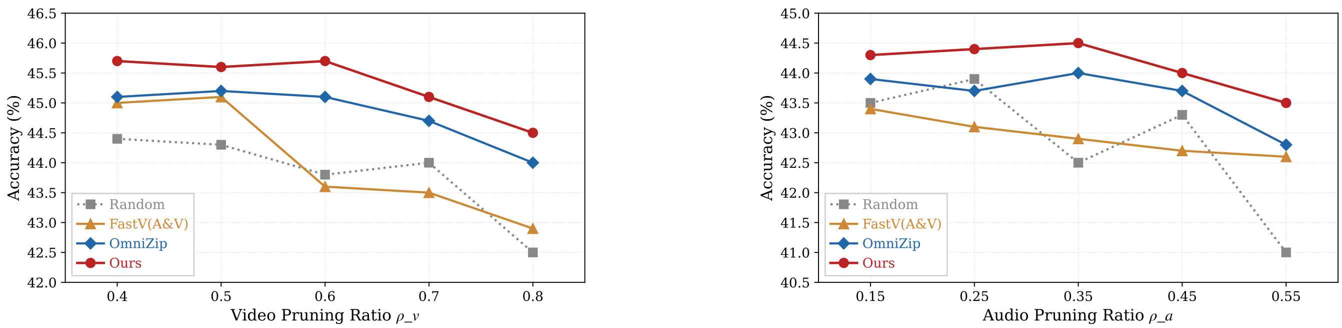


Figure 4: Ablation results for video and audio compression ratios, evaluated on Qwen2.5-Omni-3B using the WorldSense benchmark. Left: Varying the video compression ratio ρ_v with audio compression ratio $\rho_a = 0.35$. Right: Varying the audio compression ratio ρ_a with video compression ratio $\rho_v = 0.8$.

the 7B model, OmniScope incurs only a 0.35-point drop while OmniZip drops by 1.55 points. This is consistent with our analysis of cross-modal salience mismatch—when the token budget is severely limited, unidirectional cross-modal guidance is more prone to erroneously discarding information at critical moments of one modality. In contrast, OmniScope allows each modality to independently assess its own query relevance, and therefore remains robust at high compression ratios.

Robustness Across Compression Ratios. As shown in Fig. 4, we evaluate the stability of OmniScope by independently sweeping the video and audio compression ratios. On the visual side, as ρ_v increases from 0.4 to 0.8, OmniScope consistently maintains the highest accuracy with the smallest decline. On the audio side, as ρ_a increases from 0.15 to 0.55, OmniScope similarly displays the most graceful degradation curve. In contrast, OmniZip with cross-modal guidance and attention-based FastV both exhibit sharp performance drops at high compression ratios. This stability stems from the decoupled compression mechanism, which concentrates the limited token budget on the most query-relevant temporal positions, thereby preserving the most critical information within fewer tokens. This further validates that the modality-decoupled compression strategy remains robust across varying compression ratios.

4.3 Efficiency Analysis

In the QA tasks evaluated in this work, the model generates only a short option answer, making the decoding-stage KV cache and computation overhead negligible. The speedup from token compression therefore manifests primarily in the prefilling stage. As shown in Table 2, OmniScope achieves the lowest GPU memory footprint at both model scales. On the 7B model, OmniScope achieves a 2.87× prefilling speedup at 45% retention (6299 ms → 2195 ms), further increasing to 3.53× at 25% retention (6299 ms → 1784 ms), while maintaining competitive accuracy. It is worth noting that OmniZip, the current state-of-the-art omnimodal compression method, requires extracting the full attention matrix from the audio encoder to assess token importance, incurring memory overhead that grows quadratically with audio length. In contrast, OmniScope's audio scoring reuses embeddings already produced during normal inference, introducing no additional memory overhead, giving it a significant scalability advantage in long-video scenarios.

Table 2: Efficiency comparison on WorldSense. GPU Mem denotes peak GPU memory usage, Prefilling Time is the time for the prefilling stage, and E2E Lat. is the end-to-end latency.

Method	Ratio	GPU Mem (G)	Prefill (ms)	E2E Lat. (s)	Acc
Qwen2.5-Omni-7B					
Full Tokens	100%	28.31	6299.90	10.965	46.0
FastV (A&V)	45%	26.73	2242.46	4.584	45.8
OmniZip	45%	26.65	2179.73	**4.508**	45.7
OmniScope (Ours)	45%	25.14	2195.90	5.284	**46.3**
OmniScope (Ours)	25%	**24.00**	**1784.45**	4.792	45.7
Qwen2.5-Omni-3B					
Full Tokens	100%	18.04	2401.11	4.521	46.1
FastV (A&V)	45%	16.82	1814.27	4.136	44.2
OmniZip	45%	16.78	1808.51	**4.016**	45.4
OmniScope (Ours)	45%	15.20	1845.17	4.728	**46.1**
OmniScope (Ours)	25%	**14.24**	**1544.31**	4.410	44.5

The end-to-end latency of OmniScope at 45% retention is higher than FastV and OmniZip due to the one-time overhead of CLIP visual scoring, which runs outside the LLM inference pipeline. Nevertheless, compared with the uncompressed Full Token baseline, OmniScope still achieves approximately 2.1× end-to-end speedup on the 7B model (10.965 s → 5.284 s). To quantify how the fixed CLIP cost is amortized, we vary the output length from 1 to 100 tokens on Qwen2.5-Omni-7B at 45% retention and measure its share of end-to-end latency. This share decreases steadily from approximately 13% at 1 token to around 7% at 100 tokens as the generation length grows. In longer generation scenarios such as video captioning, the prefilling speedup and KV cache savings from token compression become the dominant factors, while the fixed CLIP cost is progressively amortized, making OmniScope's end-to-end latency advantage increasingly pronounced.

Table 3: Ablation on query-guided scoring. WS and DO denote WorldSense and DailyOmni, respectively (same for Tables 5 and 4). "Uniform" denotes equal compression across all positions without query-aware budget allocation. "X-guide Y" compresses Y less at positions where X is less compressed. "X-guide Y (inv.)" compresses Y more at positions where X is less compressed. All variants use Qwen2.5-Omni-3B with (ρ_v=0.6, ρ_a=0.25).

Variant	WS	DO	Avg.	Δ
Ours (full)	**46.1**	**59.3**	**52.70**	–
A&V uniform	44.5	57.6	51.05	−1.65
Audio uniform	45.7	59.0	52.35	−0.35
Visual uniform	44.7	57.6	51.15	−1.55
Audio-guide Visual (inv.)	45.2	58.2	51.70	−1.00
Visual-guide Audio (inv.)	45.6	57.8	51.70	−1.00
Audio-guide Visual	45.0	58.2	51.60	−1.10
Visual-guide Audio	45.9	58.9	52.40	−0.30

Table 4: Ablation on vision pruning strategy. All variants use Qwen2.5-Omni-3B with (ρ_v=0.6, ρ_a=0.25).

Variant	WS	DO	Avg.	Δ
Ours (full)	**46.1**	**59.3**	**52.70**	–
Delta: temporal diff only	45.6	58.7	52.15	−0.55
Delta: irreplaceability only	46.0	57.9	51.95	−0.75
All Anchor	45.2	58.0	51.60	−1.10
All Delta	45.0	57.1	51.05	−1.65

4.4 Ablation Study

Ablation Study of Query-Guided Scoring. Table 3 compares the proposed query-aware scoring with various alternative strategies. Experimental results indicate that applying uniform compression to both modalities simultaneously at the same compression ratio causes a substantial performance degradation, with an average drop of 1.65 points. Applying uniform compression to only a single modality similarly impairs performance, confirming that dynamic compression is indispensable for both modalities. Furthermore, OmniScope consistently outperforms both vision-guided audio compression and audio-guided visual compression variants. This further validates that the scores from a single modality cannot reliably predict the token importance of another modality. In contrast, decoupling and independently compressing different modalities based on query relevance more robustly preserves key information.

Ablation Study of Visual Pruning Strategy. As shown in Table 4, both the *All Anchor* and *All Delta* variants lead to noticeable performance degradation, demonstrating that simultaneously preserving temporal dynamics and spatial structural information is crucial for video understanding. Under this alternating framework, applying dynamic compression to Delta frames consistently outperforms the fixed compression strategy. This indicates the necessity of an optimal trade-off between incremental information density and global semantic information across varying compression rates. Furthermore, we conduct a sensitivity analysis on the switching threshold

Table 5: Ablation on audio compression strategy. We compare our audio compression method against mainstream compression alternatives. Qwen2.5-Omni-3B with (ρ_v=0.6, ρ_a=0.25).

Variant	WS	DO	Avg.	Δ
Ours (full)	**46.1**	**59.3**	**52.70**	–
Uniform sampling	45.7	59.2	52.45	−0.25
Average pooling	45.4	58.2	51.80	−0.90
Random drop	45.3	58.2	51.75	−0.95
Energy-based	44.1	55.8	49.95	−2.75

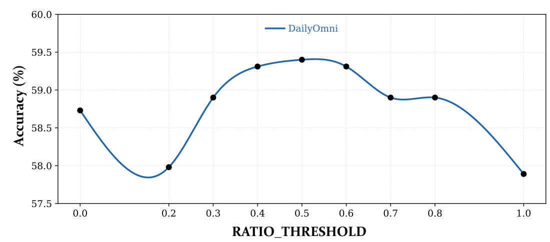


Figure 5: Effect of AD-STC's Delta-scoring threshold τ_r on DailyOmni accuracy with Qwen2.5-Omni-3B (ρ_v=0.6, ρ_a=0.25).

τ_r evaluated on the DailyOmni dataset. As shown in Fig. 5, performance exhibits an inverted U-shaped trend and peaks at an intermediate τ_r, demonstrating robustness to threshold selection.

Ablation Study of Audio Compression Strategy. As shown in Table 5, we compare our method against four mainstream alternatives. Methods that directly discard tokens (random drop, energy-based filtering) suffer the largest degradation, as they break temporal continuity and irreversibly lose information; energy-based filtering performs worst because token energy in the audio embedding space reflects acoustic loudness rather than semantic importance. Uniform sampling and average pooling preserve temporal structure but do not fully exploit the information carried by the compressed tokens. Our method merges similar tokens rather than discarding them, maximally preserving information while maintaining global temporal ordering, and therefore achieves the best performance.

5 Conclusion

We present OmniScope, a training-free, modality-decoupled token compression framework for omnimodal large language models. OmniScope uses the query as a shared semantic anchor while allowing visual and audio tokens to independently assess their importance and allocate compression budgets, avoiding the fragility of unidirectional guidance under cross-modal salience mismatch. On top of this decoupled allocation, the visual side employs anchor-delta spatio-temporal compression and the audio side adopts per-second token merging, each minimizing information loss for its respective modality. On four audio-video understanding benchmarks at two model scales, OmniScope achieves the best average performance at all compression ratios, while delivering substantial prefilling speedup and memory savings.

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